Chapter 6 Breakout (Pt. III)

STAT 4714

Questions

1. If a random variable X has the gamma distribution with $\alpha = 2$ and $\beta = 1$, find $P(1.8 < X < 2.4)$.

$$P(1.8 < X < 2.4) = P(X \leq 2.4) - P(X \leq 1.8)$$

$$= \text{pgamma}(2.4, 2, 1/1) - \text{pgamma}(1.8, 2, 1/1)$$

$$= \underline{0.1544}$$

2. Suppose that the time (in hours) required to repair a heat pump is a random variable X having a gamma distribution with $\alpha = 2$ and $\beta = 1/2$. What is the probability that, on the next service call, at least 2 hours will be required to repair the heat pump?

X = # of hours required to repair the heat pump

$P(X > 2) = 1 - P(X \leq 2)$

$1/(1/2) = 2$

$= 1 - \text{pgamma}(2, 2, 2)$

$= \underline{0.0916}$

3. The length of time to be served at a cafeteria is a random variable having an exponential distribution with a mean of 4 minutes. What is the probability that a person is served less than 3 minutes on at least 4 of the next 6 days?

$\beta = 4$

X = # of minutes until served <u>per person</u>

$P(X < 3) = P(X \leq 3)$

$= \text{pexp}(3, 1/4) = 0.5276$

Y = # of days (out of 6) served under 3 minutes

$p = 0.5276$ (from above)

$n = 6$

$P(Y \geq 4) = 1 - P(Y < 4) = 1 - P(Y \leq 3)$

$= 1 - \text{pbinom}(3, 6, 0.5276)$

$= \underline{0.3968}$

4. Suppose that the service life (in years) of a hearing aid battery is a random variable having a Weibull distribution with $\alpha = 1/2$ and $\beta = 2$. What is the probability that such a battery will be operating after 2 years?

X = service life

$$P(X>2) = 1 - P(X \leq 2)$$

$$= 1 - \text{pweibull}(2, \beta, \alpha^{-1/\beta})$$

$$= 1 - \text{pweibull}(2, 2, 1.414214)$$

$$= \underline{0.1353}$$

5. The lives of a certain automobile seal have the Weibull distribution with a failure rate $Z(t) = 1/\sqrt{t}$. Find the probability that such a seal is still intact after 4 years.

Recall that $Z(t) = \alpha\beta t^{\beta-1}$.

We need to solve for α & β to characterize our Weibull.

$Z(t) = t^{-1/2} \Rightarrow \beta = 1/2 \Rightarrow Z(t) = \alpha(1/2)t^{-1/2} \Rightarrow \alpha = 2$

With $\alpha = 2$, $\beta = 1/2$, we see $Z(t) = \alpha\beta t^{\beta-1} = t^{-1/2}$ ✓

Now, we can solve the problem!

$$P(X > 4) = 1 - P(X \leq 4)$$
$$= 1 - \text{pweibull}(4, \beta, \alpha^{-1/\beta})$$
$$= 1 - \text{pweibull}(4, 1/2, 1/4)$$
$$= \underline{0.0183}$$

6. Suppose that a study of a certain computer system reveals that the response time (in seconds) has an exponential distribution with a mean of 3 seconds. What is the probability that response time exceeds 10 seconds?

$\beta = 3$

X = length of time until response (in seconds)

$P(X > 10) = 1 - P(X \leq 10)$

$= 1 - \text{pexp}(10, 1/3)$

$= \underline{0.0357}$

7. The length of time (in seconds) that someone takes to read their email is distributed as a lognormal random variable with $\mu = 1.8$ and $\sigma^2 = 4.0$. What is the probability that they read emails for more than 20 seconds?

X = length of time (in seconds)

$P(X > 20) = 1 - P(X \leq 20)$

$= 1 - \text{plnorm}(20, 1.8, \sqrt{4})$

$= \underline{0.2750}$